# Entropic Scalar-Tensor Gravity as Brans–Dicke Theory: Planck 2018 and BBN Constraints and Prospects for LISA Standard Sirens

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**Abstract**

We study the observational status of Entropic Scalar-Tensor Gravity (ESTG), a scalar-tensor framework in which a scalar field, motivated by the local entanglement entropy density, couples to curvature through $F(\phi) = M_{\rm Pl}^2 \exp(2\beta\phi/M_{\rm Pl})$. We show that the ESTG Jordan-frame action with a canonical kinetic term is exactly Brans–Dicke theory with $\omega_{\rm BD} = 1/(4\beta^2)$ at $F = M_{\rm Pl}^2$, so that its cosmological limit is Brans–Dicke plus a cosmological constant. We also show that the commonly used effective-field-theory parameterisation $\alpha_M \propto \Omega_{\rm DE}$ with the scalar-tensor relation $\alpha_B = -\alpha_M$ is gradient-unstable for every non-zero amplitude, and is therefore not a valid description of this model. Solving the full linear Brans–Dicke cosmology with `hi_class` and fitting the Planck 2018 temperature, polarisation and lensing likelihoods together with a BBN prior on the baryon density, we compute the profile likelihood of $\beta$. We find $\beta < 0.024$ (95% one-sided), equivalent to $\omega_{\rm BD} > 421$ and to a present-day running of the Planck mass $\alpha_{M,0} < 3.7 \times 10^{-3}$. The best fit lies at $\beta = 0$; ESTG does not improve the fit ($\Delta$AIC $= +2$). Along the profile the best-fit Hubble constant increases with $\beta$, reflecting the known $H_0$–$\omega_{\rm BD}$ degeneracy. Finally, we assess LISA. A running Planck mass changes the gravitational-wave luminosity distance rather than the waveform phase, and at the Planck bound the effect is $d_L^{\rm GW}/d_L^{\rm EM} - 1 \approx 1 \times 10^{-3}$ at $z = 1$. A Fisher forecast for massive black-hole binary standard sirens gives $\sigma(\alpha_{M,0}) \approx 0.04$–$0.07$ for 10–30 events, an order of magnitude weaker than the CMB bound. LISA standard sirens are therefore not expected to improve the constraint on ESTG.

## 1 Introduction

General Relativity (GR) has achieved remarkable predictive success across a vast range of scales, from precision Solar System experiments to cosmological surveys. However, foundational puzzles—most notably the cosmological constant problem, the unknown nature of dark matter and dark energy, and the non-renormalizability of gravity—continue to motivate extensions of Einsteinian gravity. Theoretical developments inspired by black hole thermodynamics, the holographic principle and AdS/CFT duality suggest that spacetime geometry and gravitational dynamics may emerge from quantum information and entanglement entropy [1, 2, 3].

In this work we consider a scalar-tensor theory motivated by this idea, in which the local entanglement entropy density couples non-minimally to curvature. The guiding postulate is:

> *Postulate:* At each spacetime point, a scalar field $\phi(x^\mu)$ associated with matter density characterizes the local quantum entanglement entropy density, entering the gravitational field equations as a dynamical source term.

Our aims are (i) to identify precisely which known theory the resulting action corresponds to on cosmological scales, (ii) to constrain its single new parameter with the full Planck 2018

likelihoods and a BBN prior, using a Boltzmann code that solves the scalar-tensor perturbation equations, and (iii) to assess whether the Laser Interferometer Space Antenna (LISA) can improve on these constraints.

## 2 Theoretical Framework

### 2.1 Action from the Entanglement Postulate

Following the thermodynamic gravity paradigm of Jacobson [1], the vacuum entanglement entropy across a local causal horizon of area $\mathcal{A}$ obeys the Bekenstein–Hawking relation $S = \mathcal{A}/(4G\hbar)$. In the presence of matter, we parameterise the deviation of the local entanglement entropy density from its vacuum value by a scalar field $\phi$:

$$s_{\rm ent}(x^\mu) = s_0 \exp\left(\frac{2\beta\phi(x^\mu)}{M_{\rm Pl}}\right) = s_0\left[1 + \frac{2\beta\phi}{M_{\rm Pl}} + \mathcal{O}\left(\frac{\beta^2\phi^2}{M_{\rm Pl}^2}\right)\right], \tag{1}$$

where $s_0 \propto M_{\rm Pl}^2$ is the vacuum entanglement scale and $\beta$ is a dimensionless coupling. Imposing the first law of local horizon thermodynamics, $\delta Q = T\, dS_{\rm total}$, with $S_{\rm total} = S_{\rm grav} + S_{\rm ent}$, the effective gravitational coupling acquires a field dependence,

$$\frac{1}{8\pi G_{\rm eff}(\phi)} = F(\phi) = M_{\rm Pl}^2 \exp\left(\frac{2\beta\phi}{M_{\rm Pl}}\right). \tag{2}$$

Promoting this relation to a covariant field theory gives the Jordan-frame action

$$S_J = \int d^4x\sqrt{-g}\left[\frac{1}{2}F(\phi)R - \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - V(\phi)\right] + S_m(g_{\mu\nu}, \psi_m), \tag{3}$$

where $V(\phi)$ is a potential (chameleon-type, Sec. 2.4) and matter couples minimally to $g_{\mu\nu}$.

### 2.2 Equivalence with Brans–Dicke Theory

Defining the dimensionless field $\Phi \equiv F(\phi)/M_{\rm Pl}^2$, we have $(\partial\Phi)^2 = (2\beta/M_{\rm Pl})^2\Phi^2(\partial\phi)^2$, and Eq. (3) becomes

$$S_J = \int d^4x\sqrt{-g}\left\{\frac{M_{\rm Pl}^2}{2}\left[\Phi R - \frac{\omega_{\rm BD}(\Phi)}{\Phi}(\partial\Phi)^2\right] - V\right\} + S_m, \tag{4}$$

with

$$\omega_{\rm BD}(\Phi) = \frac{1}{4\beta^2\,\Phi}. \tag{5}$$

This is Brans–Dicke theory with a potential. Since $\Phi$ changes by only a few per cent between BBN and today for the couplings of interest (Sec. 3.3), we treat

$$\omega_{\rm BD} = \frac{1}{4\beta^2} \tag{6}$$

as constant, its value at $\Phi = 1$. The same result follows in the Einstein frame. With $\tilde{g}_{\mu\nu} = \Phi\, g_{\mu\nu}$, the canonical field $\varphi$ satisfies

$$\left(\frac{d\varphi}{d\phi}\right)^2 = \frac{3}{2}M_{\rm Pl}^2\left(\frac{F_{,\phi}}{F}\right)^2 + M_{\rm Pl}^2\frac{1}{F} = 6\beta^2 + e^{-2\beta\phi/M_{\rm Pl}}, \tag{7}$$

and matter couples through $A(\varphi) = \Phi^{-1/2}$ with strength

$$\alpha_\varphi \equiv M_{\rm Pl}\frac{d\ln A}{d\varphi} = -\frac{\beta}{\sqrt{6\beta^2 + e^{-2\beta\phi/M_{\rm Pl}}}} \xrightarrow{\phi=0} -\frac{\beta}{\sqrt{1+6\beta^2}}. \tag{8}$$

The standard relation $2\omega_{\rm BD}+3 = 1/(2\alpha_\varphi^2)$ [4] then reproduces Eq. (6) exactly. GR is recovered for $\beta \to 0$.

On cosmological scales we take the potential to act as a cosmological constant. This is the natural assumption for a chameleon-type potential, since such fields cannot drive cosmic acceleration in a way that differs observably from $\Lambda$ [5]. The cosmological limit of ESTG is therefore Brans–Dicke theory plus $\Lambda$, with the single new parameter $\beta$.

### 2.3 Gravitational Waves and the Running Planck Mass

In the effective-field-theory (EFT) description of dark energy [6], the theory is characterised by the effective Planck mass $M_*^2 = F$ and the functions $\alpha_K$, $\alpha_B$, $\alpha_M$, $\alpha_T$. For Eq. (3),

$$\alpha_M \equiv \frac{d\ln M_*^2}{d\ln a} = \frac{2\beta\,\dot\phi}{H M_{\rm Pl}}, \qquad \alpha_B = -\alpha_M, \qquad \alpha_T = 0, \tag{9}$$

in the convention of Ref. [6]. Tensor perturbations obey

$$\ddot h_{ij} + (3+\alpha_M) H \dot h_{ij} - \frac{1}{a^2}\nabla^2 h_{ij} = 0. \tag{10}$$

Since $\alpha_T = 0$, gravitational waves propagate at the speed of light, consistent with the joint observation of GW170817 and GRB 170817A [7, 8]. The running $\alpha_M$ only adds friction, which changes the amplitude of gravitational waves but not their phase (Sec. 4).

### 2.4 Local Tests and Screening

Without screening, Brans–Dicke theory gives the post-Newtonian parameter $\gamma - 1 = -1/(2+\omega_{\rm BD}) \simeq -4\beta^2$. The Cassini bound $|\gamma-1| < 2.3\times10^{-5}$ [9] then requires $\beta \lesssim 2.4\times10^{-3}$. Pulsar-timing tests of dipole radiation in white dwarf–neutron star binaries give constraints of comparable strength [10]. Larger couplings are viable only if the scalar is screened in the Solar System. For a chameleon [11] the Einstein-frame effective potential is

$$V_{\rm eff}(\varphi) = U(\varphi) + \tilde\rho_m A(\varphi), \qquad m_{\rm eff}^2 = U_{,\varphi\varphi} + \tilde\rho_m A_{,\varphi\varphi}, \tag{11}$$

where $U = V M_{\rm Pl}^4/F^2$. The field becomes heavy, and its fifth force short-ranged, in dense environments. Two consequences matter here. First, black holes carry no scalar charge in Brans–Dicke theory, so binary black holes emit no dipole radiation. Second, for chameleon screening to operate in the Milky Way, the scalar's Compton wavelength today must be $\lesssim 1$ Mpc, which makes its effect on linear cosmological scales negligible [5]. The cosmological constraint derived below therefore applies to the branch of the theory in which the field is light on cosmological scales. In the screened branch, CMB data are essentially insensitive to $\beta$.

## 3 Cosmological Constraints

### 3.1 Why Not $\alpha_M \propto \Omega_{\rm DE}$

A common phenomenological choice is to take all EFT functions proportional to the dark-energy fraction, $\alpha_i(a) = c_i\,\Omega_{\rm DE}(a)$, so that $\alpha_M(a) = \alpha_{M,0}\,\Omega_{\rm DE}(a)/\Omega_{\rm DE,0}$. Imposing the ESTG relations $\alpha_B = -\alpha_M$ and $\alpha_T = 0$ (with $\alpha_K = \Omega_{\rm DE}$), we find with `hi_class` that the scalar sound speed squared becomes negative for every $c_M \neq 0$. For $c_M > 0$ the gradient instability appears at early times, with $c_s^2 \simeq -c_M$. For $c_M < 0$ it appears today, with $c_s^2 \simeq c_M$. A time dependence $\alpha \propto \Omega_{\rm DE}$ is not one that a consistent Brans–Dicke background produces. We therefore solve the covariant Brans–Dicke theory directly instead of using an EFT parameterisation.

### 3.2 Method and Data

We use `hi_class` [12, 13] (version 3.4.0.1, based on CLASS v3.4 [14, 15]) with its covariant `brans_dicke` model. Its parameters are $(V_0, \omega_{\rm BD}, \Phi_{\rm ini}, \Phi'_{\rm ini})$. $V_0$ is fixed by the closure relation and $\Phi_{\rm ini}$ by requiring $M_*^2 = 1$ today, i.e. the cosmological Newton constant equals its laboratory value. The field starts on its radiation-era attractor, and $\omega_{\rm BD}$ is set by Eq. (6). The present-day running $\alpha_{M,0}$ is read off the background solution. Numerically we find $\alpha_{M,0} \simeq 6.2\,\beta^2$ for a Planck-like cosmology. The likelihood is evaluated with Cobaya [16] and the following data:

1. **Planck 2018 CMB** [17, 18]: low-$\ell$ TT and low-$\ell$ EE, the high-$\ell$ TT,TE,EE `plik_lite` likelihood (marginalised over foregrounds, with the calibration $A_{\rm Planck}$ free), all in their native Python implementations, and the CMB lensing likelihood [19].

2. **BBN**: a Gaussian prior $\omega_b = 0.0222 \pm 0.0005$, a conservative width that encompasses primordial deuterium determinations [20].

We follow the Planck baseline with one massive neutrino of 0.06 eV, and use HMcode [21] for the non-linear matter power spectrum entering the lensing likelihood. The free parameters are $\{\ln(10^{10}A_s), n_s, H_0, \omega_b, \omega_{\rm cdm}, \tau_{\rm reio}, A_{\rm Planck}\}$ plus $\beta$. As a consistency check, at $\beta = 0$ the Brans–Dicke model and a pure $\Lambda$CDM run through the same code agree to $\Delta\chi^2 = 0.016$ in total.

### 3.3 Profile Likelihood

Because $\beta$ is the only parameter beyond $\Lambda$CDM, we constrain it with a profile likelihood. For each value of $\beta$ on a grid we hold $\beta$ fixed and maximise the likelihood (times the BBN and calibration priors) over all other parameters with Cobaya's derivative-free minimiser [22], starting from the previous grid point's best fit. The result is shown in Table 1 and Fig. 1. At $\beta = 0$ the best fit is the standard Planck $\Lambda$CDM cosmology ($H_0 = 67.35$, $n_s = 0.965$), with $\chi^2 = 23.3$ (low-$\ell$ TT), 396.1 (low-$\ell$ EE), 582.8 (`plik_lite`) and 8.8 (lensing).

Table 1: Profile likelihood of $\beta$ from Planck 2018 + BBN. $\Delta\chi^2$ is relative to the global minimum, which lies at $\beta = 0$. $\alpha_{M,0}$ is the present-day running of the Planck mass from the `hi_class` background, and $H_0$ is the best-fit value in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

| $\beta$ | $\omega_{\rm BD}$ | $\alpha_{M,0}$ | $H_0$ | $\Delta\chi^2$ |
|---|---|---|---|---|
| 0 | $\infty$ | 0 | 67.35 | 0.00 |
| 0.004 | 15625 | $9.9 \times 10^{-5}$ | 67.41 | 0.01 |
| 0.008 | 3906 | $4.0 \times 10^{-4}$ | 67.59 | 0.13 |
| 0.012 | 1736 | $8.9 \times 10^{-4}$ | 67.71 | 0.56 |
| 0.016 | 977 | $1.6 \times 10^{-3}$ | 68.17 | 1.03 |
| 0.020 | 625 | $2.5 \times 10^{-3}$ | 68.72 | 1.73 |
| 0.025 | 400 | $3.9 \times 10^{-3}$ | 69.62 | 2.85 |
| 0.030 | 278 | $5.7 \times 10^{-3}$ | 70.96 | 4.37 |
| 0.040 | 156 | $1.0 \times 10^{-2}$ | 73.64 | 10.48 |

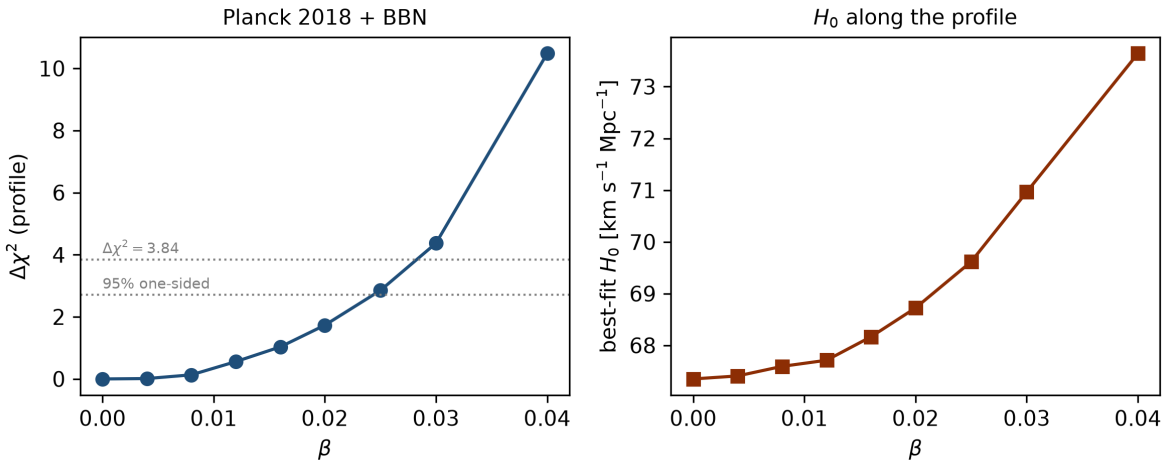



Figure 1: Left: profile likelihood $\Delta\chi^2(\beta)$ from Planck 2018 (low-$\ell$ TT, low-$\ell$ EE, high-$\ell$ `plik_lite` TT,TE,EE, lensing) and a BBN prior on $\omega_b$. Dotted lines mark $\Delta\chi^2 = 2.71$ (95% one-sided) and 3.84. Right: best-fit $H_0$ along the profile.

Since $\beta \geq 0$ and the minimum lies on the boundary $\beta = 0$, we quote the one-sided 95% limit from $\Delta\chi^2 = 2.71$, obtained by linear interpolation between grid points:

$$\beta < 0.024 \quad \Longleftrightarrow \quad \omega_{\rm BD} > 421 \quad \Longleftrightarrow \quad \alpha_{M,0} < 3.7 \times 10^{-3} \qquad (95\%). \tag{12}$$

The more conservative $\Delta\chi^2 = 3.84$ threshold gives $\beta < 0.028$ ($\omega_{\rm BD} > 313$). The numerical accuracy of the individual minimisations is $\delta\chi^2 \approx 0.1$–$0.2$, which translates into an uncertainty of about $\pm 0.001$ on the limit. This bound is of the same order as previous CMB constraints on Brans–Dicke theory [23, 24]. It is about an order of magnitude weaker than the unscreened Solar System bound of Sec. 2.4, but it probes the scalar on cosmological scales and at early times.

At the 95% limit the effective gravitational constant during BBN, $G \propto 1/M_*^2$, exceeds its present value by 1.9%. This is well within BBN bounds on a varying $G$, which are at the $\sim 10\%$ level [25]. In our pipeline this shift enters the expansion history but not the CLASS helium-abundance prediction, which is a negligible effect at this level.

### 3.4 Model Comparison

The global minimum of the profile is at $\beta = 0$, so adding $\beta$ does not lower $\chi^2$: $\Delta\chi^2 = \chi^2_{\rm ESTG} - \chi^2_{\rm GR} = 0$. The Akaike Information Criterion therefore changes by

$$\Delta{\rm AIC} = \Delta\chi^2 + 2\,\Delta k = +2, \tag{13}$$

mildly in favour of GR. Current CMB and BBN data show no preference for ESTG. A Bayesian evidence calculation requires a full posterior and depends on the prior range assumed for $\beta$; we leave it for future work.

The right panel of Fig. 1 shows that the best-fit $H_0$ rises with $\beta$, from 67.4 at $\beta = 0$ to 69.6 near the 95% limit, because the larger effective gravitational constant in the early universe ($M_*^2 < 1$ before today) raises the early expansion rate and shrinks the sound horizon [24]. With CMB data alone this direction is only weakly penalised. Adding baryon acoustic oscillation data would test whether ESTG can relieve the Hubble tension without spoiling the fit to the expansion history.

# 4 Prospects for LISA

## 4.1 LISA Sensitivity

We model the sky-averaged LISA sensitivity with the analytic curve of Robson, Cornish & Liu [26], with arm length $L = 2.5 \times 10^9$ m and transfer frequency $f_* = c/(2\pi L) \approx 19.09$ mHz:

$$S_n(f) = \frac{10}{3L^2}\left[P_{\rm oms}(f) + \frac{4P_{\rm acc}(f)}{(2\pi f)^4}\right]\left[1 + \frac{6}{10}\left(\frac{f}{f_*}\right)^2\right] + S_c(f), \tag{14}$$

where

$$P_{\rm oms}(f) = (1.5 \times 10^{-11}\ {\rm m})^2\left[1 + \left(\frac{2\ {\rm mHz}}{f}\right)^4\right], \tag{15}$$

$$P_{\rm acc}(f) = (3.0 \times 10^{-15}\ {\rm m/s^2})^2\left[1 + \left(\frac{0.4\ {\rm mHz}}{f}\right)^2\right]\left[1 + \left(\frac{f}{8\ {\rm mHz}}\right)^4\right], \tag{16}$$

and $S_c(f)$ is the Galactic confusion noise for a 4-year mission. Figure 2 shows that massive black-hole binaries (MBHBs) at $z = 1$ are detected with signal-to-noise ratios of order $10^2$–$10^3$.

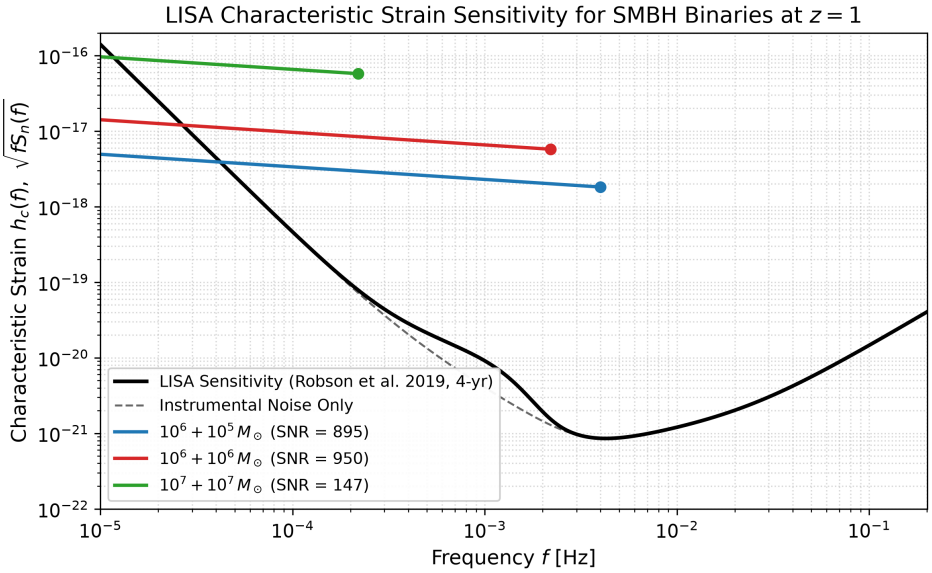



Figure 2: Sky-averaged LISA characteristic strain sensitivity $\sqrt{fS_n(f)}$ [26] (solid: including 4-year Galactic confusion noise; dashed: instrumental noise only), compared with the characteristic inspiral strain $h_c(f) = 2f|\tilde{h}(f)|$ of MBHBs at $z = 1$. Filled circles mark the ISCO frequencies.

## 4.2 The Waveform Phase Does Not Constrain $\alpha_M$

Equation (10) shows that $\alpha_M$ enters only as friction, so it changes the amplitude of the signal during propagation but not its phase. Two further points rule out a phase test. First, black holes carry no scalar charge in Brans–Dicke theory, so there is no dipole term. Second, any correction proportional to the leading Newtonian phase, $\delta\Psi \propto (\pi\mathcal{M}_c f)^{-5/3}$ plus a constant, is an exact linear combination of the derivatives of $\Psi$ with respect to $\ln\mathcal{M}_c$ and $\phi_c$. It is absorbed into the chirp mass and coalescence phase.

We verified numerically that a Fisher analysis over $\{\ln\mathcal{M}_c, t_c, \phi_c, \delta\}$ with such a term does return a finite $\sigma(\delta)$. That value, however, comes entirely from the $\mathcal{M}_c^{5/6}$ scaling of the amplitude:

it equals $[(5/6)\,\mathrm{SNR}\,k]^{-1}$ with $k = (2/3)\pi^{5/3}$. Once the luminosity distance is included as a free parameter, the Fisher matrix becomes singular to machine precision. Such phase-based estimates should therefore not be used for this class of theories [27].

### 4.3 Standard Sirens

The observable effect of a running Planck mass is the ratio of the gravitational-wave and electromagnetic luminosity distances [28, 29]:

$$\frac{d_L^{\mathrm{GW}}(z)}{d_L^{\mathrm{EM}}(z)} = \frac{M_*(0)}{M_*(z)} = \exp\left[\frac{1}{2}\int_0^z \frac{\alpha_M(z')}{1+z'}\,dz'\right]. \tag{17}$$

We compute $M_*^2(z)$ directly from the `hi_class` Brans–Dicke background. In the regime of interest the deviation is linear in $\alpha_{M,0}$: at the 95% Planck limit, $d_L^{\mathrm{GW}}/d_L^{\mathrm{EM}} - 1 = 0.7$, 1.1, 1.6 and $2.3 \times 10^{-3}$ at $z = 0.5$, 1, 2 and 4.

To measure this ratio, sirens need an electromagnetic counterpart that provides the redshift. We forecast $\sigma(\alpha_{M,0})$ for $N$ MBHB sirens spread uniformly over $0.5 \le z \le 4$. For each siren the fractional distance error combines three terms in quadrature:

- an instrumental error (5% baseline, 2% optimistic);
- weak lensing, $\sigma_{\mathrm{lens}}/d_L = 0.066\,[(1-(1+z)^{-0.25})/0.25]^{1.8}$ [30], either without delensing (baseline) or reduced by 50% (optimistic);
- peculiar velocities of 500 $\mathrm{km\,s^{-1}}$ [31].

$H_0$ and $\Omega_m$ are marginalised with Planck-like Gaussian priors. The results are given in Table 2.

Table 2: Forecast $1\sigma$ error on $\alpha_{M,0}$ from $N$ LISA MBHB standard sirens with electromagnetic counterparts ($0.5 \le z \le 4$), for the Brans–Dicke evolution of $M_*(z)$. The Planck + BBN 95% limit is $\alpha_{M,0} < 3.7 \times 10^{-3}$.

| $N$ | 1 | 3 | 10 | 30 | 100 |
|---|---|---|---|---|---|
| Baseline (5% instr., no delensing) | 0.18 | 0.12 | 0.065 | 0.041 | 0.029 |
| Optimistic (2% instr., 50% delensing) | 0.088 | 0.061 | 0.036 | 0.027 | 0.022 |

Over a 4-year mission LISA is expected to observe a few to a few tens of MBHB sirens with counterparts, depending on the population model [30]. Even in the optimistic case, $\sigma(\alpha_{M,0}) \gtrsim 0.03$, roughly ten times larger than the Planck bound. The forecast saturates at large $N$ because of the $H_0$ and $\Omega_m$ priors. LISA standard sirens alone are therefore not expected to improve the constraint on ESTG.

## 5 Discussion and Conclusions

Our main results are:

- **ESTG is Brans–Dicke theory.** With a canonical kinetic term, the coupling $F(\phi) = M_{\mathrm{Pl}}^2 e^{2\beta\phi/M_{\mathrm{Pl}}}$ is equivalent to Brans–Dicke theory with $\omega_{\mathrm{BD}} = 1/(4\beta^2)$. Its cosmological limit is Brans–Dicke + $\Lambda$, and it satisfies $c_T = c$.
- **The $\alpha_M \propto \Omega_{\mathrm{DE}}$ parameterisation is unstable.** Combined with the scalar-tensor relation $\alpha_B = -\alpha_M$, it is gradient-unstable for every non-zero amplitude, so constraints derived from it do not apply to ESTG.

- **Cosmological bound.** A profile likelihood with the full Planck 2018 likelihoods and a BBN prior gives $\beta < 0.024$, i.e. $\omega_{\rm BD} > 421$ and $\alpha_{M,0} < 3.7 \times 10^{-3}$ (95%). The best fit is GR ($\Delta\text{AIC} = +2$). The best-fit $H_0$ grows with $\beta$.
- **LISA.** The waveform phase does not constrain $\alpha_M$. Standard sirens reach $\sigma(\alpha_{M,0}) \approx 0.03$–$0.07$, an order of magnitude weaker than the CMB bound.

Several caveats apply. The cosmological bound assumes that the scalar is light on cosmological scales. If chameleon screening is active in the Milky Way, the field's cosmological effect is negligible [5], and the relevant constraints are local and depend on the screening potential. We treated $\omega_{\rm BD}$ as constant, neglecting its few-per-cent variation through $\Phi$. The lensing likelihood uses HMcode, which is calibrated on $\Lambda$CDM. Natural extensions are a full Bayesian MCMC, the addition of BAO and supernova data to test the $H_0$ direction, and a treatment of the screened regime.

## Data Availability

The analysis pipeline (Cobaya configuration, `hi_class` interface, profile-likelihood and standard-siren codes) and the best-fit files for every grid point are available at https://github.com/batu-ta/ESTG_MCMC_LISA in the `analysis/` directory. The results use `hi_class` 3.4.0.1 (commit `b425aef`) and Cobaya 3.6.2.

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